

Overview of Optimization for ML

Sunita Sarawagi

Motivation

Chap 8 PML book: $L(w) \rightarrow F(w)$
 $L(\theta)$

- Many ML training algorithms can be posed as a continuous optimization problem

$$\min_{\substack{\{w_1, \dots, w_d\} \\ w_j \in \mathbb{R}}} F(w_1, w_2, \dots, w_d) \Leftrightarrow \min_{\substack{w \\ w \in \mathbb{R}^d}} F(w)$$

where the variables are the parameters of the model

$F(w)$: loss over training data + regularizer

- Loss = continuous function representing error.
- Finding the minima of general functions could be intractably hard
 - Doable for certain types of functions $\rightarrow$ convex functions.

Matrix Calculus

Gradient and Hessian of multi-variate scalar functions.

$$f(x) = x^3 + \log x \quad \left| \right. \quad \frac{\partial f}{\partial x} = 3x^2 + \frac{1}{x}$$

Calculus of function over multi-dimension variables.

$$F(w_1, w_2, \dots, w_d) \equiv F(\underline{w})$$

Gradient of F :

$\nabla_w F(w) =$

$d \times 1$

$$\begin{bmatrix} \frac{\partial F}{\partial w_1} \\ \frac{\partial F}{\partial w_2} \\ \vdots \\ \frac{\partial F}{\partial w_d} \end{bmatrix}$$

Gradient at a w^*

$$\nabla_w F(w) \big|_{w=w^*}$$

$$\nabla_{w^*} F(w^*)$$

Double derivative of a univariate function.

$$f(x) = x^3 + \log x$$

$$f'(x) = \frac{\partial f(x)}{\partial x} = 3x^2 + \frac{1}{x} \quad ; \quad f''(x) = \frac{\partial^2 f(x)}{\partial x^2} = 6x - \frac{1}{x^2}$$

Multivariate functions: double derivative is called Hessian of the function.
 $F(w)$ which is twice differentiable w.r.t all w_i, w_j
then its Hessian

$$H_{F(w)} = \nabla^2 F = \begin{bmatrix} \frac{\partial^2 F}{\partial w_1^2} & \frac{\partial^2 F}{\partial w_1 \partial w_j} & \dots & \frac{\partial^2 F}{\partial w_1 \partial w_d} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\partial^2 F}{\partial w_i \partial w_1} & \frac{\partial^2 F}{\partial w_i \partial w_j} & \dots & \frac{\partial^2 F}{\partial w_i \partial w_d} \\ \vdots & \vdots & \vdots & \ddots \\ \frac{\partial^2 F}{\partial w_d \partial w_1} & \frac{\partial^2 F}{\partial w_d \partial w_j} & \dots & \frac{\partial^2 F}{\partial w_d \partial w_d} \end{bmatrix} d \times d$$

Example

$$F(w_1, w_2) = (w_1 + 10)^2 + 5w_2^2 + w_1 w_2$$

$$\nabla_w F|_{w=[1,1]} \Rightarrow \begin{bmatrix} 2(w_1 + 10) + w_2 \\ 10w_2 + w_1 \end{bmatrix}_{(1,1)} =$$

$$\begin{bmatrix} 23 \\ 11 \end{bmatrix}$$

$$H_F : w=[1,1] = \begin{bmatrix} 2 & 1 \\ 1 & 10 \end{bmatrix}$$

Gradients of common functions.

$$\textcircled{1} F(w) = \mathbf{a}^T \mathbf{w} = \sum_{j=1}^d a_j w_j \quad \mathbf{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix}$$

$$\nabla_w F = \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} = \mathbf{a}$$

$$\boxed{\nabla_w (\mathbf{a}^T \mathbf{w}) = \mathbf{a}}$$

$$\textcircled{2} F(w) = \underbrace{\mathbf{b}^T}_{d \times 1} \underbrace{\mathbf{A}}_{d \times d} \mathbf{w} = \sum_{i=1}^d \sum_{j=1}^d b_i A_{ij} w_j$$

$$\boxed{\nabla_w (\mathbf{b}^T \mathbf{A} \mathbf{w}) = \mathbf{A}^T \mathbf{b}}$$

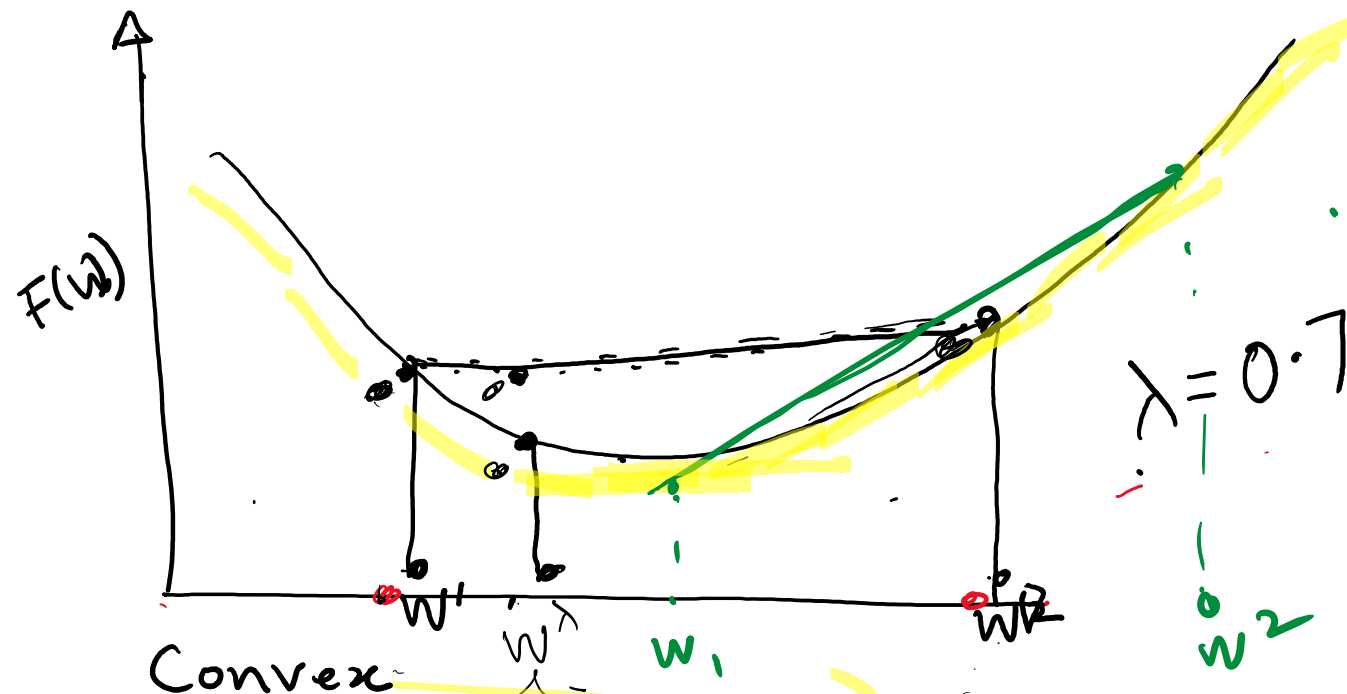
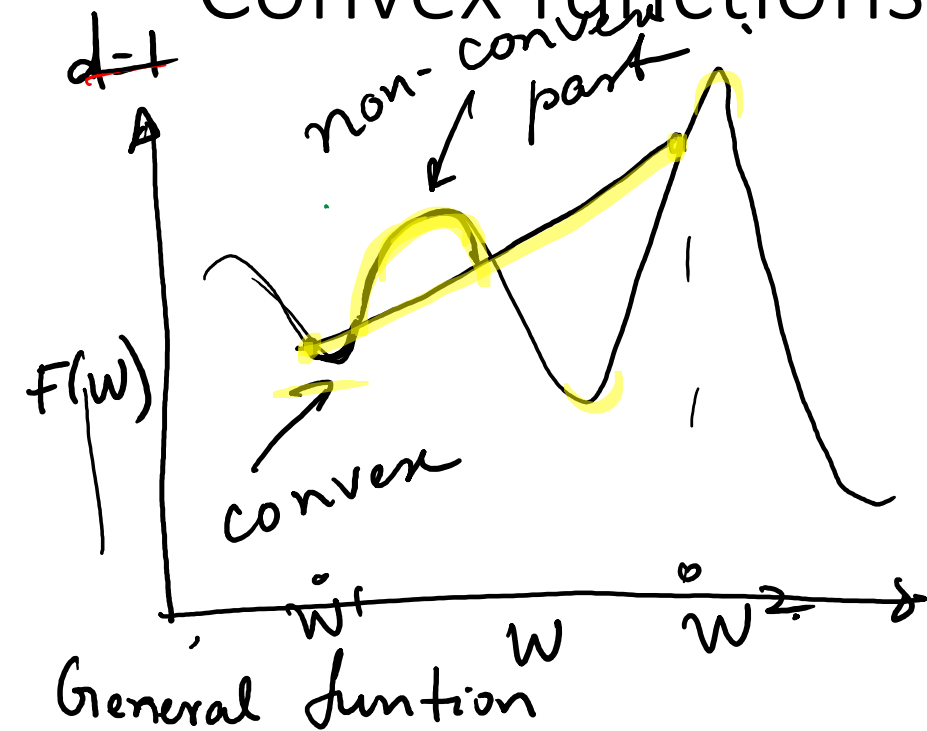
$$\nabla_w F = \mathbf{A}^T \mathbf{b} \quad \sum_{i=1}^d \sum_{j=1}^d w_i A_{ij} w_j$$

$$\textcircled{3} F(w) = \underbrace{\mathbf{w}^T \mathbf{A} \mathbf{w}}_{\substack{\rightarrow \\ \text{not linear in } w}} =$$

$$\nabla_w F = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}$$

$$\boxed{\nabla_w (\underbrace{\mathbf{w}^T \mathbf{A} \mathbf{w}}_{\substack{\uparrow \\ \text{quadratic form}}}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}}$$

Convex functions



A function $F(w)$ is convex in w if and only if (iff) for any $w^1, w^2 \in \mathbb{R}^d, \lambda \in [0, 1]$,

$$F(\lambda w^1 + (1 - \lambda)w^2) \leq \lambda F(w^1) + (1 - \lambda)F(w^2)$$

$F(w^\lambda)$

Concave functions = negative of convex functions.

Testing for convexity: Univariate functions

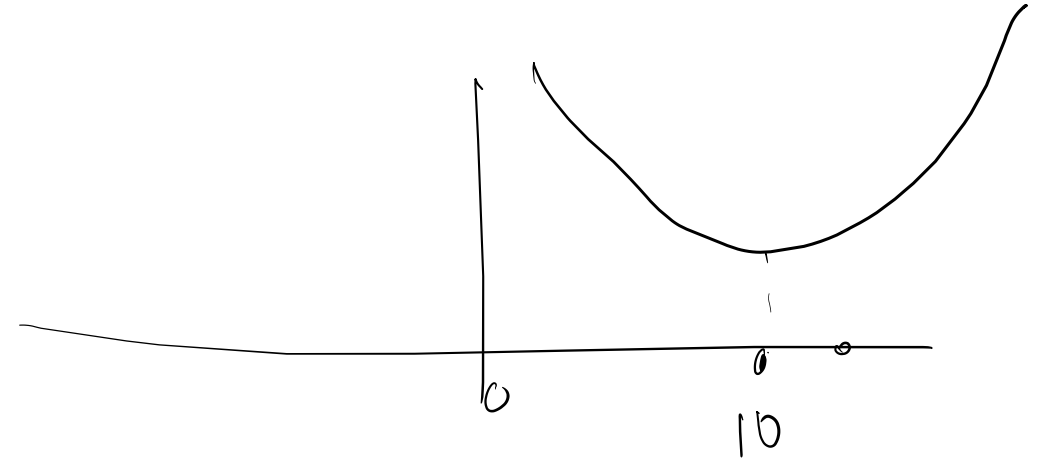
- For convex differentiable functions, we intuitively know that the gradient always increases with w . This implies that the rate of change of gradient is positive. In other words, the double derivative is greater never negative.

$$\frac{\partial^2 F(w_1)}{\partial w_1^2} \geq 0$$

Example

- Show that $(w - 10)^2$ is convex in w .

$$\frac{\partial^2 f}{\partial w^2} = 2 \geq 0$$



Example

$$\underline{F(w)} = -\sum_{i=1}^N k_i \log w + w$$

$$\frac{\partial^2 F(w)}{\partial w^2} = \frac{\partial}{\partial w} \left[-\sum_{i=1}^N \frac{k_i}{w} + 1 \right]$$

$$= \sum_{i=1}^N k_i / w^2$$

$$\sum_{i=1}^N k_i \geq 0$$

$F(w)$ is convex if

Testing for convexity: multi-variate functions

A function $F(w)$; $w \in \mathbb{R}^d$ is convex iff its Hessian is positive semi-definite at all w .

$H(w)$ ← matrix of size $d \times d$
 $\xrightarrow{F} \mathbb{R}$ should be positive semi-definite (psd)
 $w^T H_F w \geq 0 \quad \forall w \in \mathbb{R}^d$

Linear Regression Training objective is convex

$$L(w) = w^T X^T X w - 2 w^T X^T y$$

$$H_L = X^T X$$

Need to show that H_L is psd

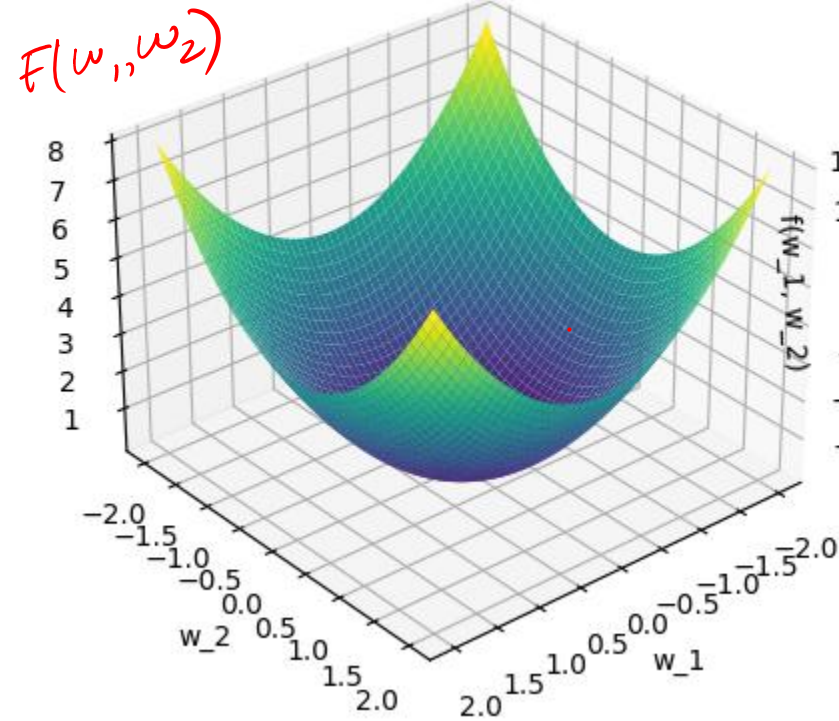
$$a^T H_L a \geq 0 \quad \forall a$$

$$\Rightarrow a^T X^T X a \geq 0 \Rightarrow$$

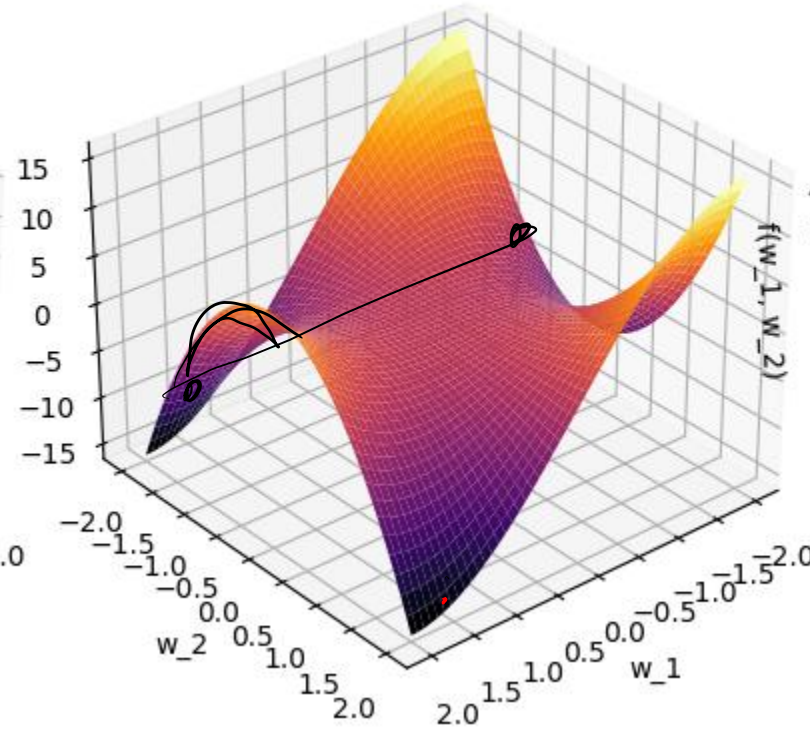
$$\underbrace{(Xa)^T}_{\vec{b}} (Xa) = \vec{b}^T \vec{b} = \|\vec{b}\|^2 \geq 0$$

Convex Vs Non-convex functions in 2-d

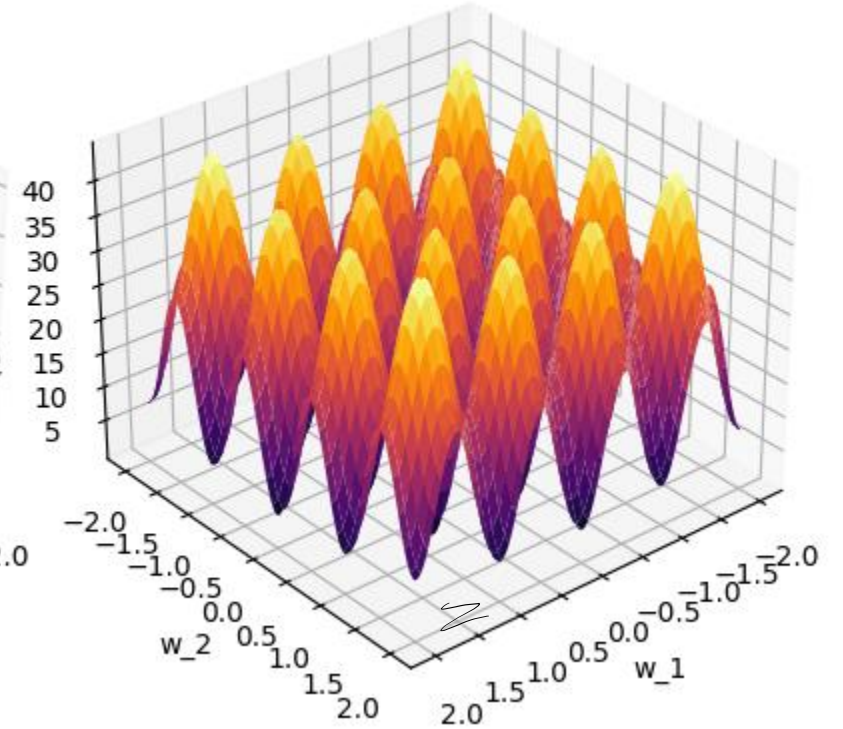
Convex Function: $f(w_1, w_2) = w_1^2 + w_2^2$



$f(w_1, w_2) = w_1^3 - 3w_1w_2^2$

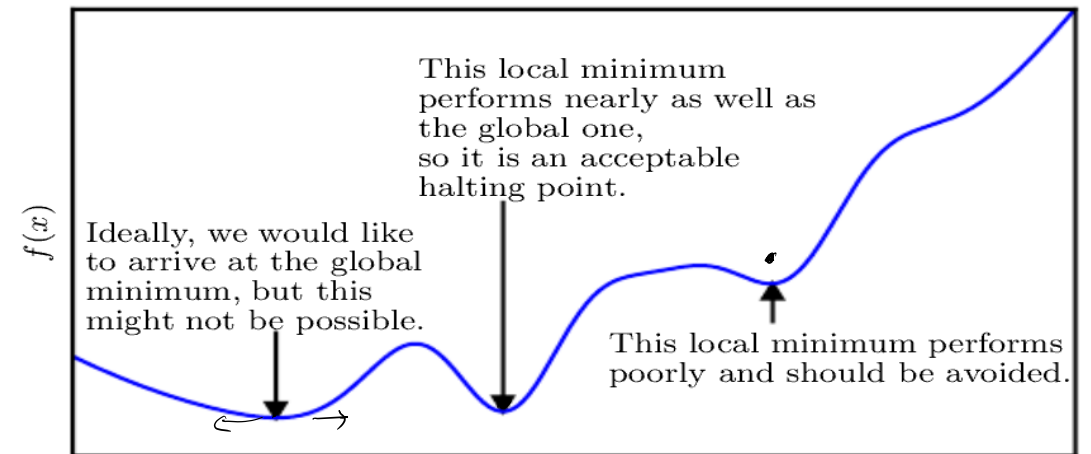
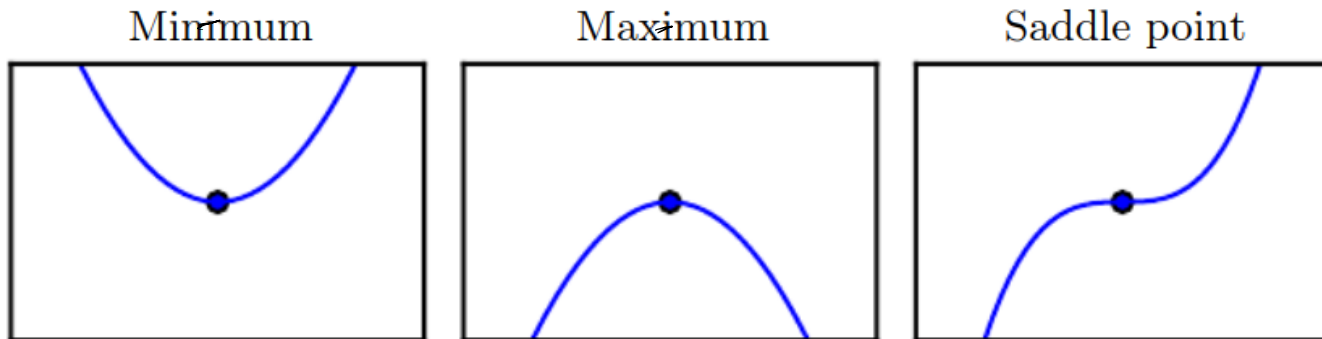


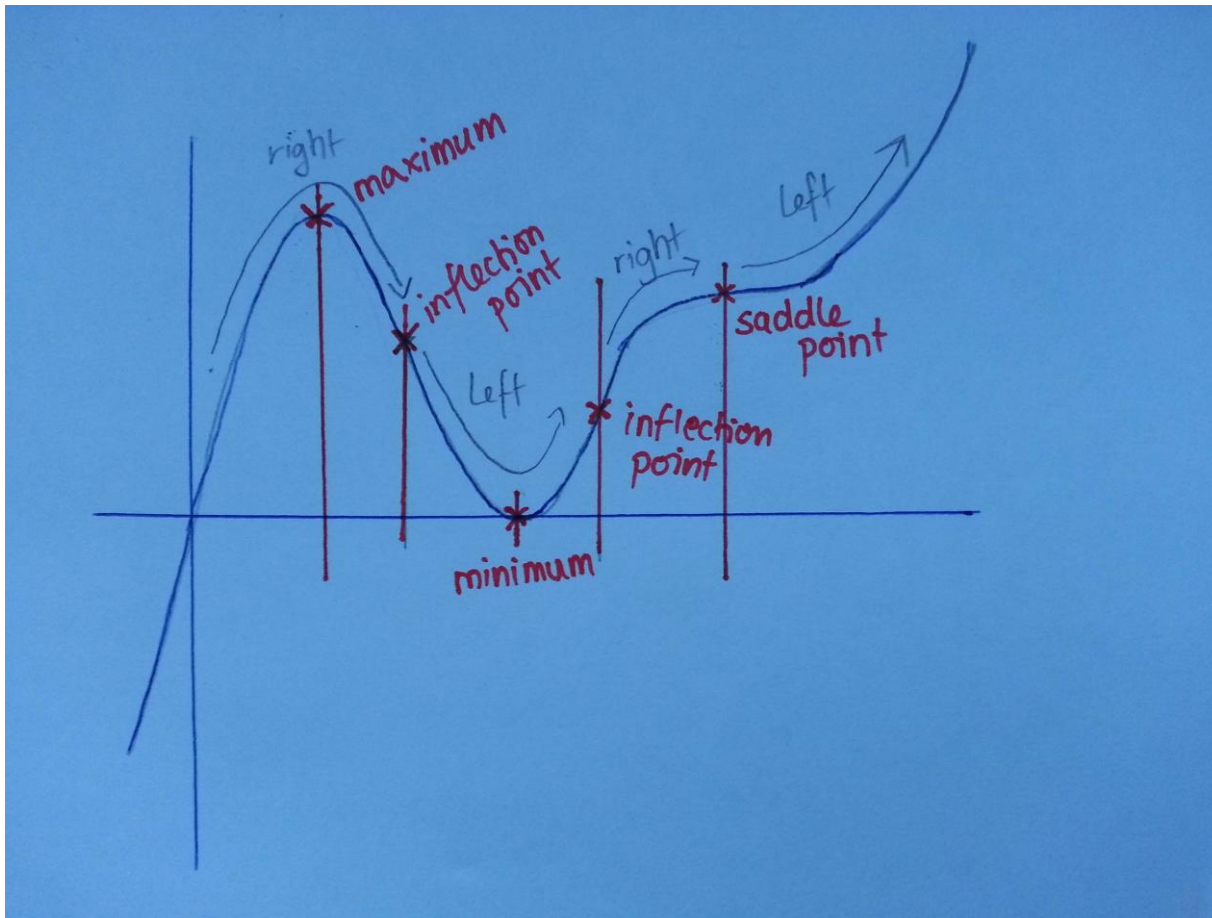
$f(w_1, w_2) = 20 + w_1^2 + w_2^2 - 10 * (\cos(2\pi w_1) + \cos(2\pi w_2))$



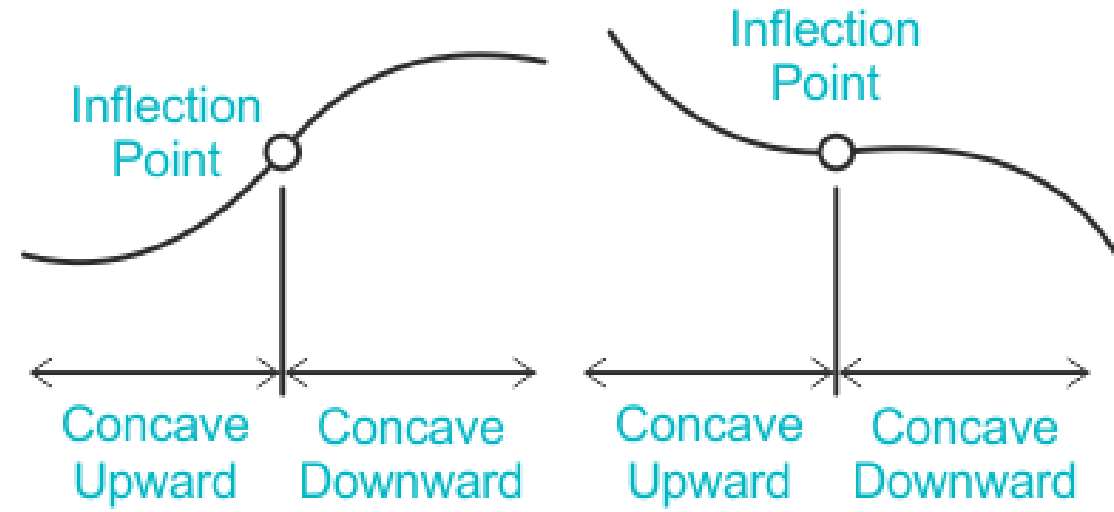
Optimizing a function

- $\min_w F(w), w \in R^d$
- A point w^* is a **global minima** of $F(w)$ if $F(w) \geq F(w^*) \quad \forall w$
- A point w^* is a **local minima** of $F(w)$ if there exists an $\epsilon > 0$ such that $F(w) \geq F(w^*) \quad \forall |w - w^*| \leq \epsilon$
- A point with zero gradient but which is not a local minima or maxima is a **saddle point**.
- A point where double derivative is 0 is an inflection point (function changes concavity)

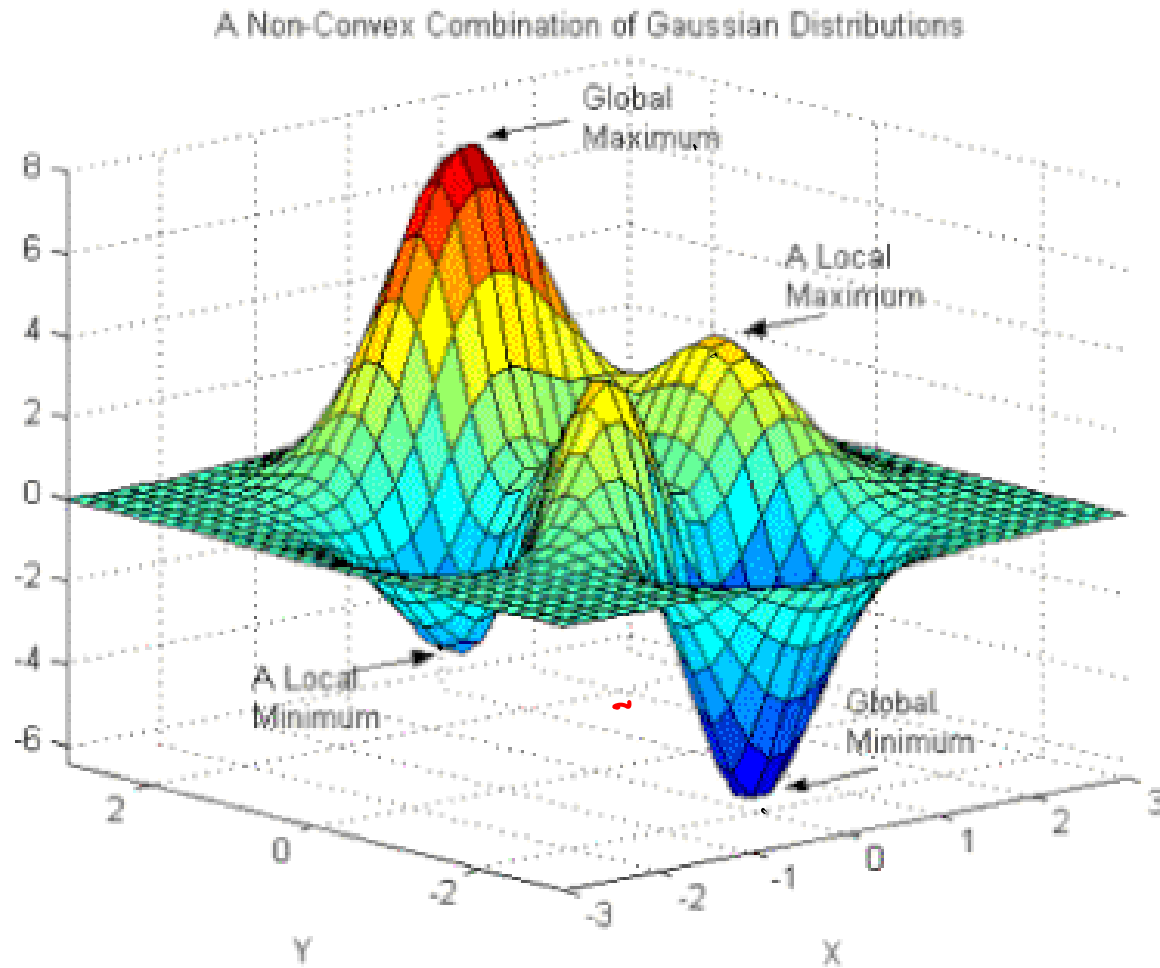




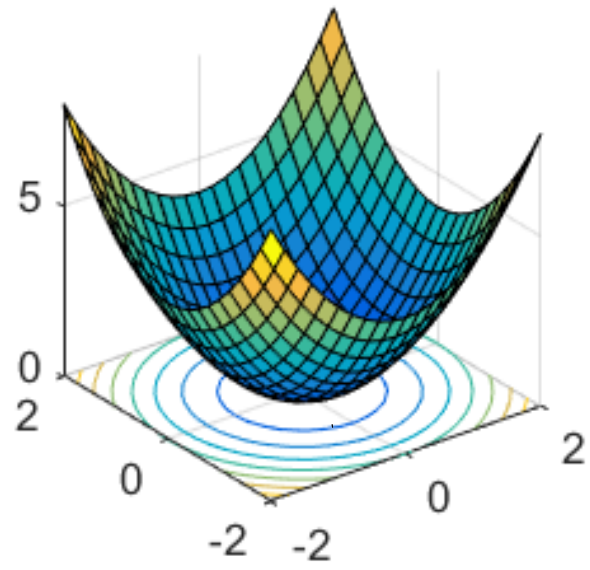
testbook



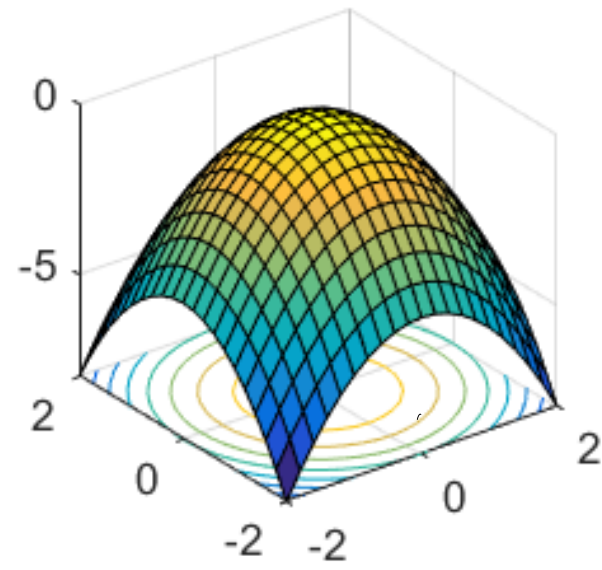
Local and global minima in 2-D



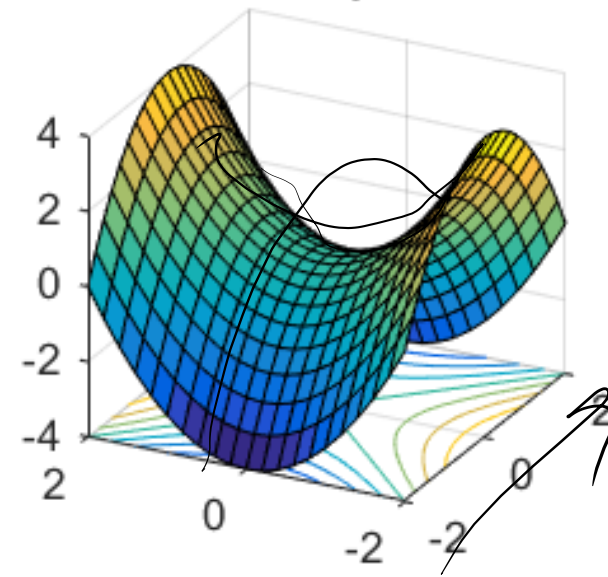
local min



local max



saddle point



Gradient of a differentiable function

- Derivative of a function on a single variable measures the rate of change of the function.

$$F(w_1) = w_1^2 + 2w_1^3$$

$$F(w_1) = \log(1 + e^{-w_1})$$

$$\frac{\partial F(w_1)}{\partial w_1} = 2w_1 + 6w_1^2$$

$$\frac{\partial F(w_1)}{\partial w_1} = \frac{-e^{-w_1}}{1 + e^{-w_1}}$$

$$F(w_1) = \sum_{i=1}^N (y^i - w_1 x^i)^2$$

$$\frac{\partial F(w_1)}{\partial w_1} = \sum_{i=1}^N 2(y^i - w_1 x^i)(-x^i)$$

Gradients of multi-variable functions

- For multivariable functions $F(w_1, \dots, w_d)$ we can define partial derivative of F w.r.t each of the variables.

$$\frac{\partial F}{\partial w_1}, \frac{\partial F}{\partial w_2}, \dots, \frac{\partial F}{\partial w_d}$$

- Gradient of $F(w)$ denoted as $\nabla F(w)$: vector of partial derivative of the function

$$\nabla_w F(\vec{w}) = \begin{bmatrix} \frac{\partial F}{\partial w_1} \\ \frac{\partial F}{\partial w_2} \\ \vdots \\ \frac{\partial F}{\partial w_d} \end{bmatrix}$$

$d=2$ Example of gradient

$$F(w_1, w_2) = F(\mathbf{w}) = \frac{1}{2} (w_1^2 + 10 w_2^2)$$

Minima and gradients

- **Theorem:** If $F(\underline{w})$ is differentiable and if $\underline{w}^*$ is a local minima of $F(\underline{w})$, then $\underline{\nabla F(w^*)} = \underline{0}$

- Theorem: For convex functions a $\underline{w}^*$ is a global minima if and only if $\underline{\nabla F(w^*)} = 0$. That is the local minima is the global minima.

Already applied this theorem to find the parameters of ordinary least square in closed form.

Iterative optimization algorithms

- Most often we will not be able to solve for $\nabla F(w) = 0$ in closed form
 - E.g. MLE for the logistic loss. (Show) *arises in training classification models*
- Iterative algorithms (General template)
 - w^0 = Choose an initial point.
 - For $t = 1$ to stopping criteria (local minima, maximum iterations, etc)
 - $w^{t+1} \leftarrow$ Move to a near-by point w^t such that $F(w)$ reduces. $F(w^{t+1}) < F(w^t)$
- Many iterative algorithms have been proposed for such cases
 - Zero-th order algorithm (line-search) for optimizing 1-d convex functions.
 - First-order or gradient-based algorithms
 - **Gradient descent**
 - Conjugate gradient descent,
 - .
 - Second-order algorithms
 - Semi or pseudo second order algorithms

Example of solving in closed form

$$F(w_1, w_2) = (w_1 - 1)^2 + 3(w_2 - 10)^2$$

$$\nabla F(w) = \begin{bmatrix} 2(w_1 - 1) \\ 6(w_2 - 10) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow 2w_1^* - 2 &= 0 & \Rightarrow w_1^* &= 1 \\ 6w_2^* - 60 &= 0 & \Rightarrow w_2^* &= 10 \end{aligned}$$

$$\begin{aligned} F(w_1, w_2) &= \log(e^{-w_1 + 2w_2}) \\ &+ \log(e^{w_1 + 7w_2}) \end{aligned}$$

First Order Taylor Expansion of Differentiable Functions

Differentiable functions resemble linear functions in a small neighborhood around each point.

- Let $\underline{w}^0 \in R^d$, $\underline{w} \in R^d$, and $\underline{F(w)}$ be a differential function. Then

$$F(w) = F(w^0) + \underline{\nabla F(w^0)^T} (w - w^0) + ||w - w^0||^2 \alpha(w^0, w - w^0)$$

Where $\underline{\alpha(w^0, w - w^0)} \rightarrow \underline{0}$ as $w \rightarrow \underline{w^0}$

$F(w) \approx F(w^0) + \nabla F(w^0)^T (w - w^0)$ whenever $||w - w^0||$ is small.
linear in w .

Example: First-order Taylor Approximation